

Reference Materials:

- 1. Kittel C., Introduction to Solid State Physics Wiley & Sons (NY) 1996.**
- 2. Tester J.W., Thermodynamics and Its Applications Prentice Hall (NJ) 1997.**
- 3. West A.R., Solid State Chemistry and Its Applications Wiley & Sons (NY) 1984.**
- 4. Venables J., Introduction to Surface and Thin Film Processes Cambridge University Press (UK) 2000.**
- 5. Website sponsored by the UK Surface Analysis Forum (USAF)
<http://www.siu.edu/~cafs/surface> (written by D.T. Marx at Southern Illinois University)**

Lecture Topics (Part 1)---

What is a Surface?

A surface is locust of points which classify the boundary between an object and its surroundings.

How are Surfaces Classified?

Surfaces are classified by the spacing between surface atoms and the # of bonds each surface atom forms with either other surface atoms or atoms in the bulk.

Why are surface atoms important?

Properties are typically based upon bulk (internal) measurements. We typically are concerned with solids having densities of 10^{23} atoms/cm³. Most of those atoms are within the solid. However, surfaces become important when we move into nanoscience--- where many more of those atoms are surface atoms. Or when phenomena of interest only occurs at the surface.

What are some of the Properties of surfaces?

The surface properties we will consider include atomic density, surface tension, and surface energy.

What are the broad categories of surface reactions we will consider?

The surface reactions we will consider are sublimation (release of atoms from a solid surface), adsorption (uptake of atoms onto a solid surface), epitaxial growth.

Lecture Topics (Part 2)---

What are the classifications of Surface Characterization?

Surfaces may be characterized with respect to their topography (i.e. roughness), chemistry, surface orientation, and thickness of chemically homogeneous regions at the surface.

Which Surface Characterization Techniques will we learn about in this lecture?

- X-ray and Neutron Reflectivity
- X-ray Photoelectron Spectroscopy
- Secondary Ion Mass Spectrometry
- Scanning Auger Microscopy

What are some other Surface Characterization Techniques of practical importance in research?

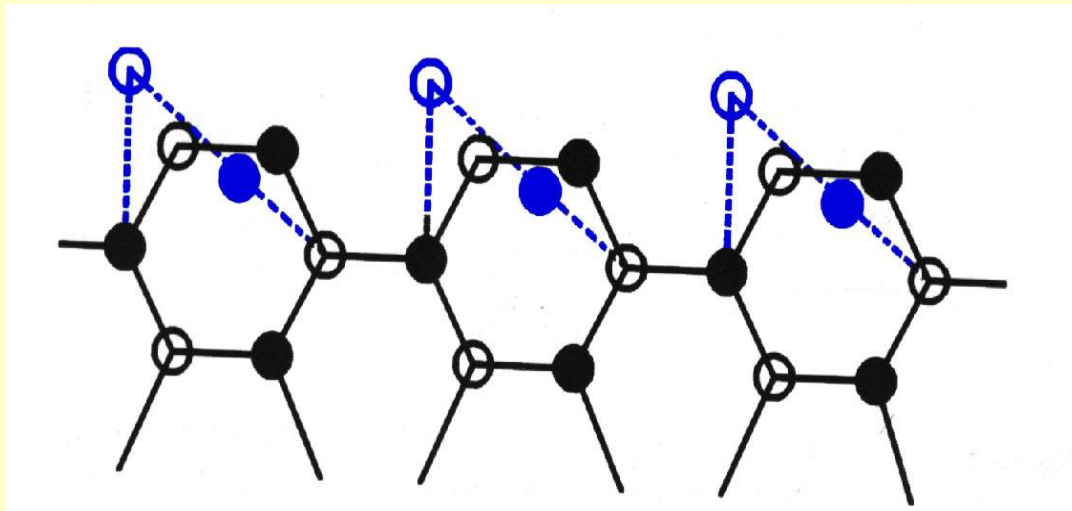
- Atomic Force Spectroscopy
- Scanning Tunneling Microscopy
- Near-IR Spectroscopy

What is a Surface?

Surfaces are defined by 'relaxed' atoms (i.e. not constrained in 3D as their internal counterparts are). Dangling bonds from these surface atoms are free to react.

Relaxation of surface atoms leads to reconstruction (rearrangement of atoms near the surface).

Relaxation and reconstruction are strongly influenced by the bonding type in the bulk material (i.e. metallic, covalent, ionic, and vander waals)



(110) Surface of GaAs. Surface atoms (blue) are 'relaxed' (i.e. not constrained in 3D).

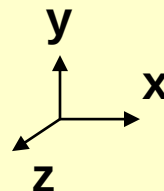
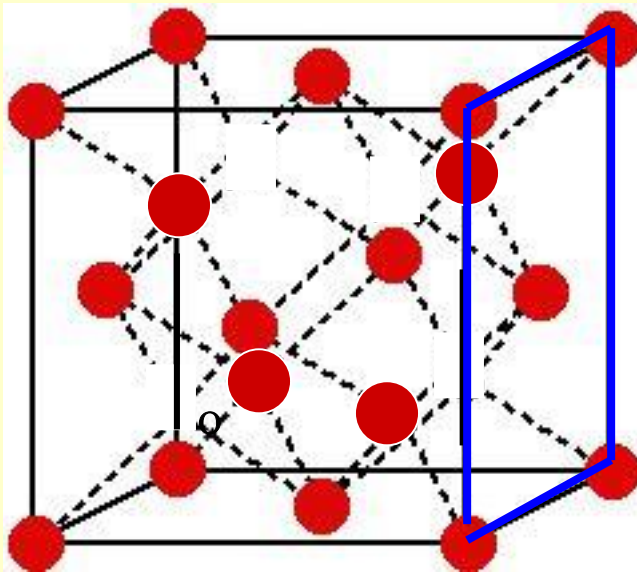
What is a Surface?

--- Review of Miller Indices in Crystallography

Miller Indices are used to identify the surface terminating.

If it is stated that the surface is a Si(100), that indicates that we are examining a surface which has 1 surface atoms spaced at $\frac{\sqrt{2}}{2}a$.

Silicon Crystal Structure



- (100) is the set of planes (hkl) which intersect the crystal at $(1/1, 1/0, 1/0)$ or $(1, \infty, \infty)$

This plane has
 $4(1/8) + 1(1/2) = 1$ Atom

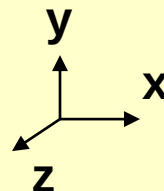
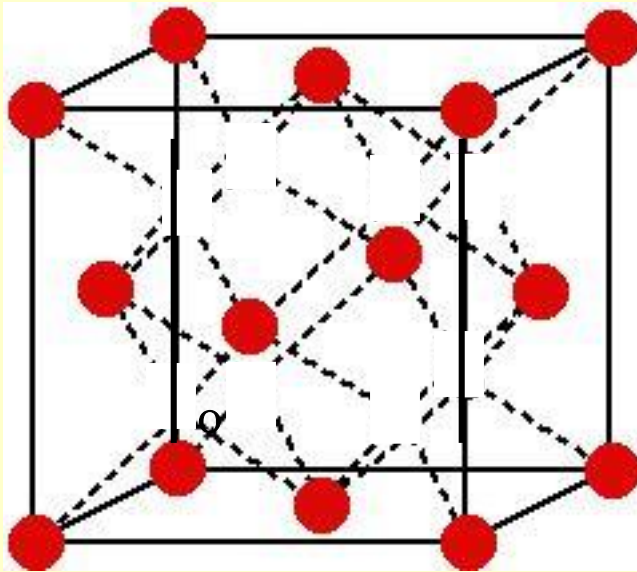
Activity

--- Using Miller Indices to Define Terminating Surface

- What is the number of atoms and atomic spacing for a surface terminating at the (110) plane of a fcc crystal?

HINT: Recall (100) is the set of planes (hkl) which intersect the crystal at $(1/1, 1/0, 1/0)$ or $(1, \infty, \infty)$

FCC Crystal Structure

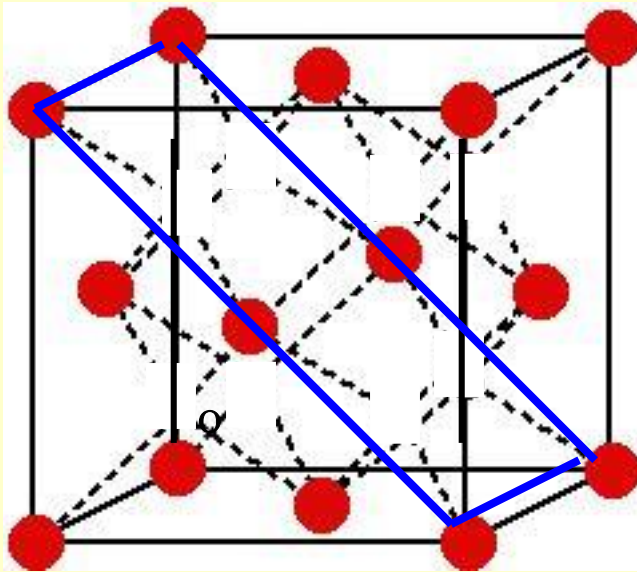


Activity

--- Using Miller Indices to Define Terminating Surface

- What is the number of atoms and atomic spacing for a surface terminating at the (110) plane of a fcc crystal?

FCC Crystal Structure

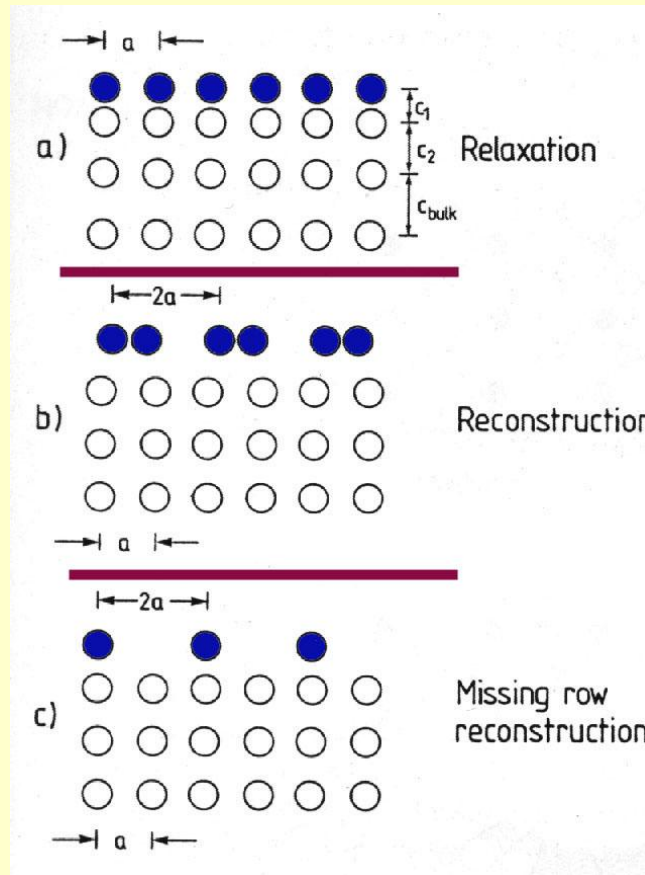


Ans: Recall (110) is the set of planes (hkl) which intersect the crystal at $(1/1, 1/1, 1/0)$ or $(1, 1, \infty)$

This plane has
 $4(1/8) + 2(1/2) = 1.5$ Atoms

Atoms are spaced at: $\frac{\sqrt{2}}{2}a$

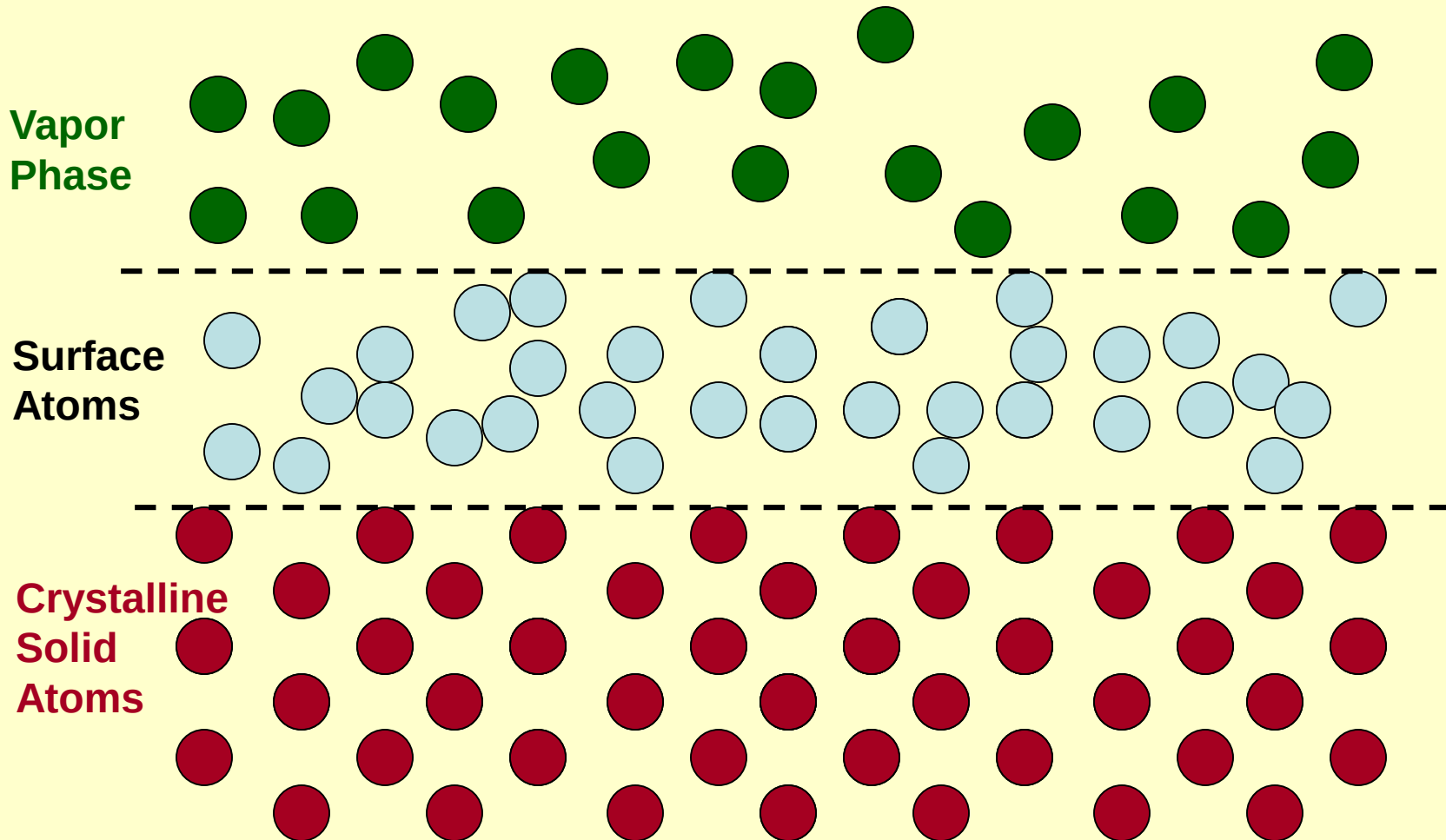
What is a Surface?



Surface atoms (blue) undergo relaxation followed by reconstruction. (a) relaxation – loss in periodic order in c direction. (b) reconstruction 1 – classified by change in atomic spacing in a direction. (c) reconstruction 2 – classified by missing row of atoms.

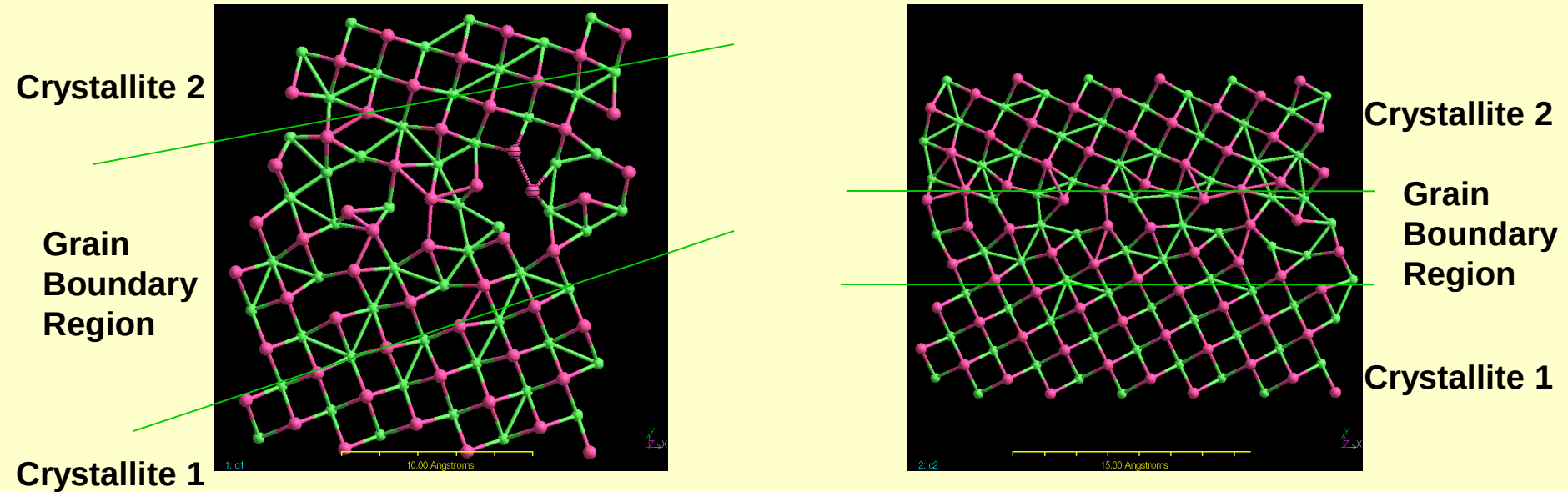
What is a Surface?

Surface atoms realize a loss in crystalline order. These atoms take on a 'pseudo' random configuration. Such a noncrystalline structure is known as 'amorphous'.



What is a Surface?

--- Grain Boundary (surface between two isomorphous solids)



Surface Atoms typically have a structure intermediate between the two terminal phases with the exception of noted solid-solid surfaces – called grain boundaries

Images taken from website – Visualization of (210) and (310) Grain Boundaries:
http://www.sv.vt.edu/classes/ESM4714/Student_Proj/class95/mutasa/mutasa.html

What is a Surface?

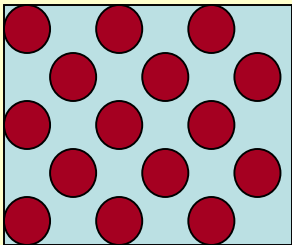
Surfaces exist between two phases.

The phases may be :

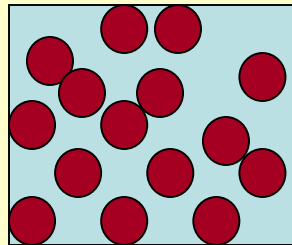
- two solids
- a solid and a vapor
- a liquid and a vapor
- a solid and a liquid

In all cases, there is a finite length of region for which the atomic packing/structure undergoes changes. The atoms in this region are the 'surface atoms'.

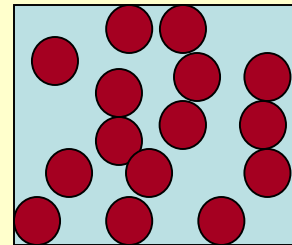
Often the surface atoms have a structure intermediate between the two terminal phases (with one noted exception: solid-solid surface/interface)



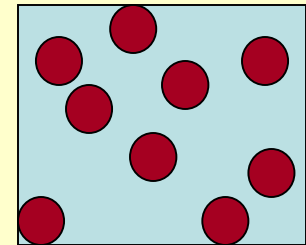
Crystalline Solid
Characterized by
Long-range Order



Amorphous Solid
Characterized by
Short-range Order



Liquid
Characterized by
Short-range Order and
rigorous atomic vibration



Vapor
Characterized by
No Order

Surface Thermodynamics

--- Treatment of Quantitative Surface Parameters

Thermodynamics is the field of science which deals with the motion of atoms under the influence of thermal driving forces.

Thermodynamic Potentials (i.e. Internal Energy (U), Helmholtz Free Energy (F), and Gibbs Free Energy (G)) have additional contribution due to surface atoms.

Contribution due to bulk atoms:

$$dF_{\text{bulk}} = -SdT - PdV + \mu dN = 0 \text{ at constant } T, V, \text{ and } N.$$

Total Free Energy contains additional contribution due surface atoms:

$$dF_{\text{Total}} = dF_{\text{bulk}} + f_s dA$$

$$dF_{\text{Total}} = f_s dA \quad \text{at constant } T, V, \text{ and } N.$$

$$dF_{\text{Total}} = \mu dN + f_s dA \quad \text{at constant } T \text{ and } V.$$

f_s is the surface excess free energy

Surface Thermodynamics

--- Surface Tension and Surface Energy

Surface Tension, γ , is the reversible work done (dW) in creating a unit area of new surface (dA).

$$\gamma = dW/dA = (dF_{\text{total}}/dA)_{T,V}$$

Since $dF_{\text{Total}} = \mu dN + f_s dA$ at constant T and V .

$$\gamma dA = \mu dN + f_s dA$$

where μ is the chemical potential of the atoms and N is the number of atoms in the system

Rewriting $\gamma = -\mu\Gamma + f_s$ at constant T and V .
where $\Gamma = -dN/dA$ (and dN always negative (-))

Conclusion: Addition of atoms/molecules to the surface (increasing N ---that is, N goes to higher negative value) region will decrease the surface tension (γ) via increase in Γ (Γ is proportional to $-dN$).

Example 1: A soap film lowers the surface tension of water because the soap molecules form monolayers at the water surface with their 'hydrophobic' ends pointing out into the gaseous regions.

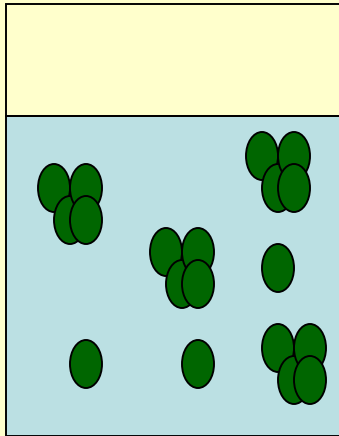
Surface Thermodynamics

--- Surface Tension and Surface Energy

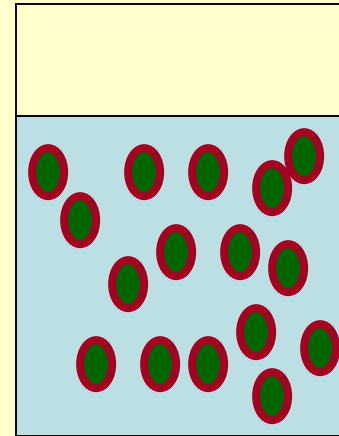
Example 2: Surfactant (polymer molecule with **hydrophobic** end group and **hydrophilic** end group) is added to nanoparticulate suspensions in order to decrease the driving force for particle agglomeration.

$$\gamma = \mu\Gamma + f_s$$

where $\Gamma = -dN/dA$



Having High Surface Energy, γ , nanoparticles will Aggregate to reduce their surface area (A)



Addition of **Surfactant** to nanoparticulate surfaces will increase N, thus decreasing surface energy, γ .
No need for aggregation to occur.

Surface Thermodynamics

--- Wulff Theorem and Surface Energy

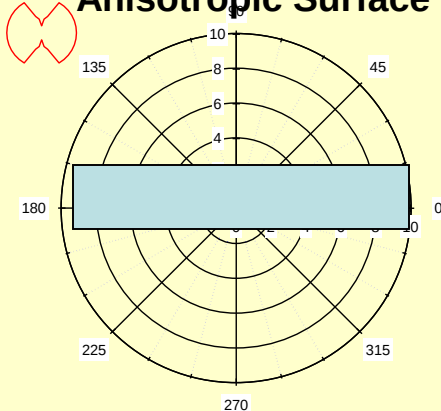
Wulff Theorem tells us that the equilibrium crystallite shape has surface planes of minimum surface energy, γ ,

Using **Wulff Construction**, we can determine the equilibrium shape of crystallites given only $\gamma(hkl)$ (i.e. surface energy for given (hkl) miller indices).

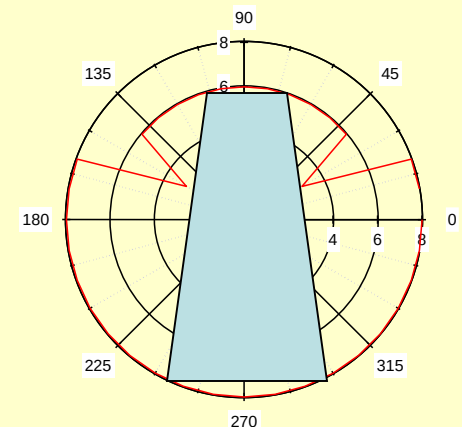
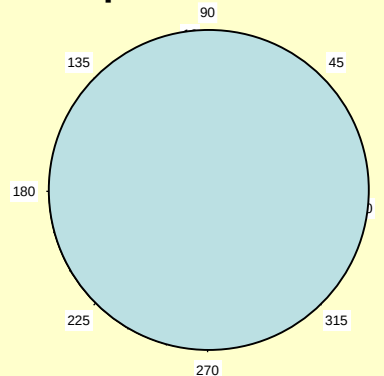
- Steps:
1. Plot polar diagram of $\gamma(\theta)$.
 2. Take the inner envelope of this diagram to get equilibrium shape.

Examples

Anisotropic Surface Energy



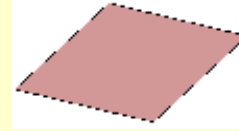
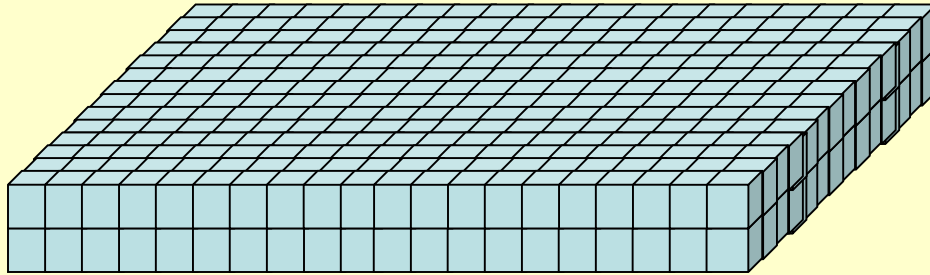
Isotropic Surface Energy



Wulff theorem applies to InGaAs quantum dot structures ---
where we may have pyramidal shapes grown from the vapor phase.

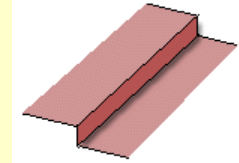
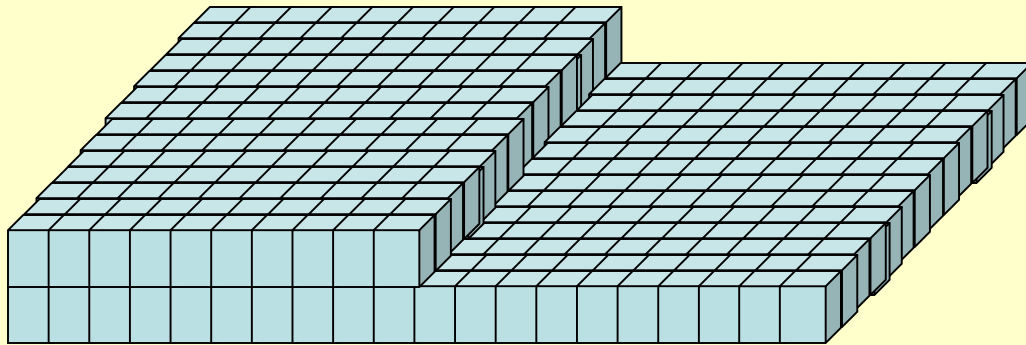
Classification of Surfaces by their Defects (or Imperfections)

---Terrace, Ledges, Kinks and Adatoms



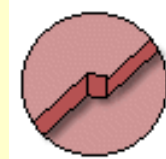
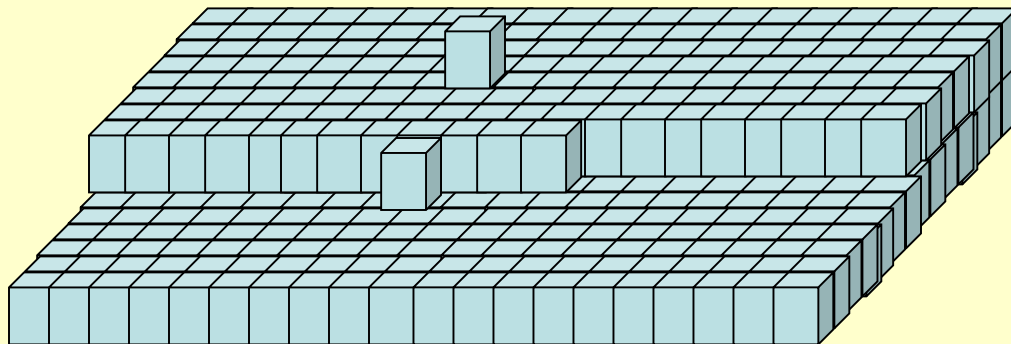
Terrace

- Surface having crystalline order



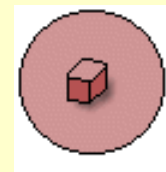
Ledge

- Steps formed at the border of terraces



Kink

- Defect formed at the end of ledges

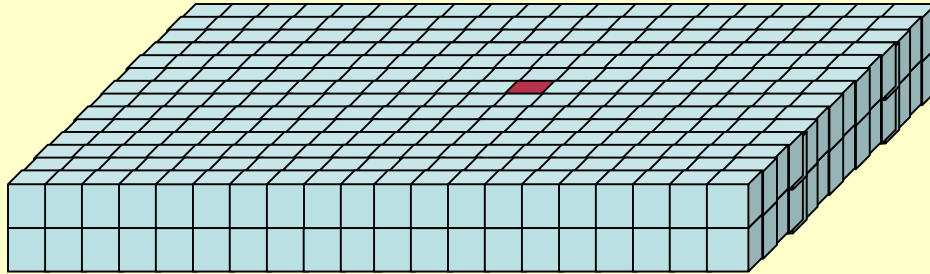


Adatom

- Single Atom sitting on a terrace or ledge surface

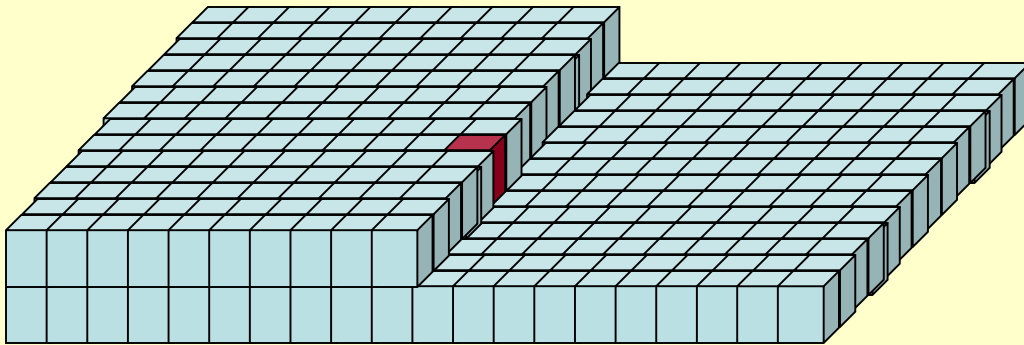
Surface Defects (or Imperfections)

---Terrace, Ledges, Kinks and Adatoms



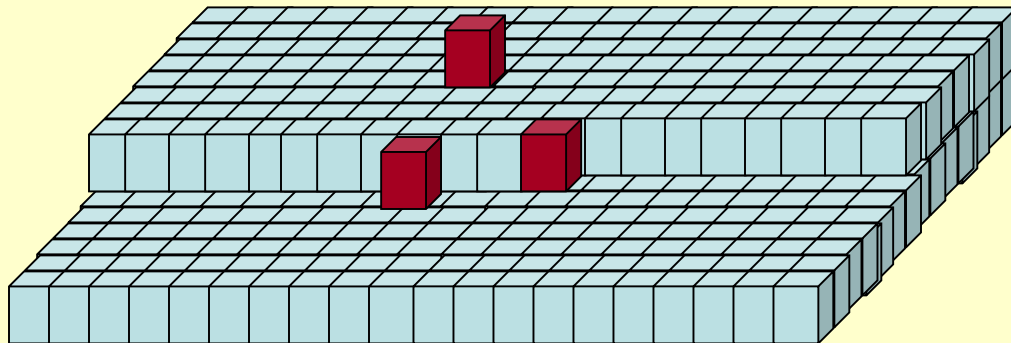
Terrace

- Terrace atom has 5 nearest neighbors



Ledge

- Ledge atom has 4 nearest neighbors



Kink

- Kink atom has 3 nearest neighbors

Adatom

- Ledge adatom has 2 nearest neighbor
- Terrace adatom has 1 nearest neighbor

Thermodynamics of Surface Defects

---Terrace, Ledges, Kinks and Adatoms

Gibbs Free Energy Equation for atom transition from terrace to ledge position.

$$\Delta G = W_{\text{ledge}} - W_{\text{terrace}}$$

W_{terrace} – Energy required to break 4 bonds.

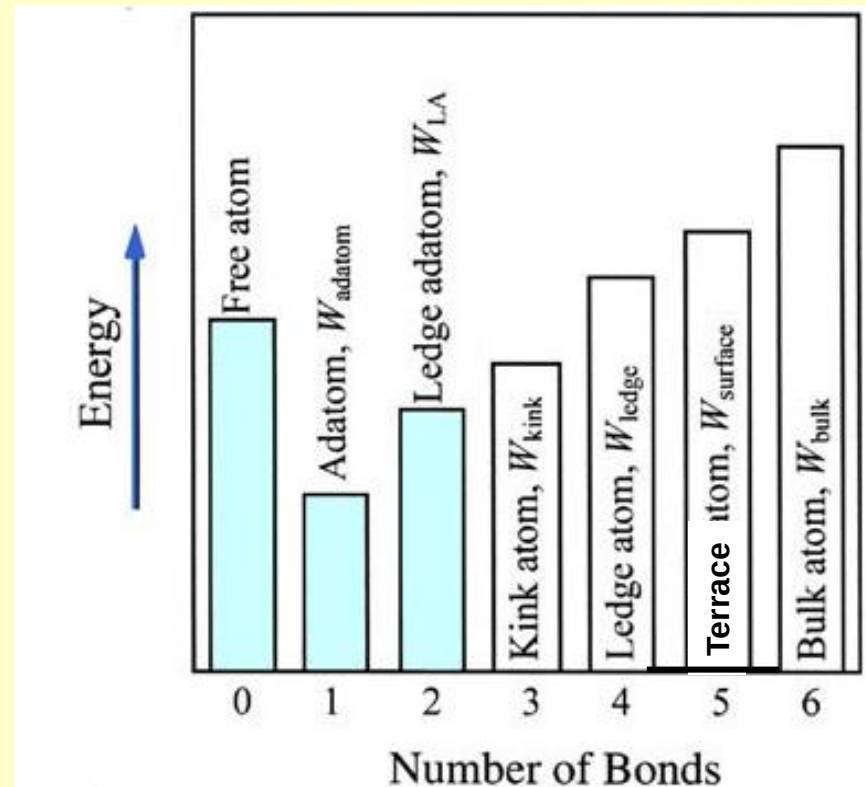
W_{ledge} – Energy required to form 5 bonds.

Arrhenius Equation represents the Temperature dependence on # of atoms undergoing transition (n).

$$n = N \exp(-\Delta G/kT)$$

where N is the # of atoms available to participate in the transition

Binding Energy for Atoms at Various Sites



Site Stability has direct proportionality to binding energy. The higher the binding energy, the higher the site stability.

Surface and their Properties

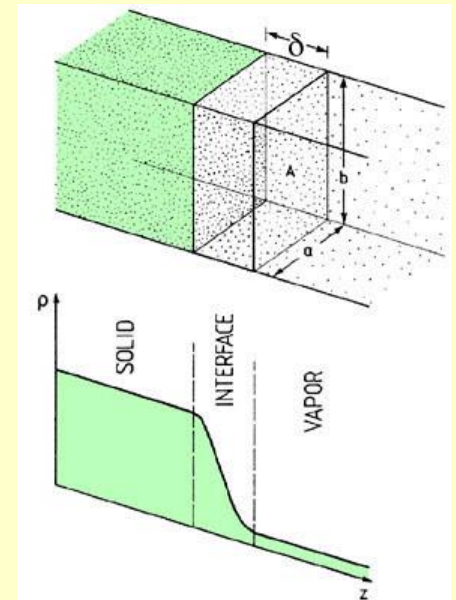
---Pressures and Forces

Gradient Physical Properties (density, etc.) between solid and vapor phase.

Force on plane $b\delta$ is $F = Pb\delta - b\gamma$

Where γ is surface tension

surface tension, γ , is the reversible work done in creating a unit area of new surface.



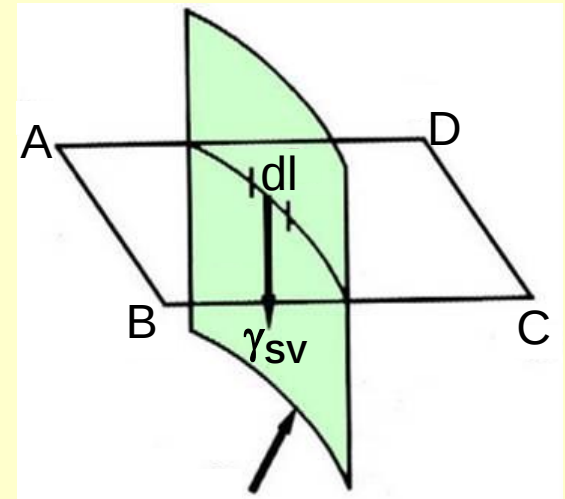
Solid-Vapor Interface
at Equilibrium.
Interfacial Area, A
Interfacial Thickness, δ

Surfaces and their Properties

---Surface Stress, γ_{sv}

Young Model (developed for Liquid Surfaces)

Tension Surface (an infinitesimally thin elastic membrane) occurs at the interface. The sum of forces acting on the length of the interfacial curve are zero. This force along a unit length (dl) of the curvature surface is known as the surface stress γ_{sv} reported in units of [N/m]. This surface stress may be reduced by increasing the length between bonds on the surface.



Tension Surface

Surfaces and their Properties

---Other Concepts Using Surface Stress, γ_{SV}

Other Concepts using Surface Stress, γ_{SV}

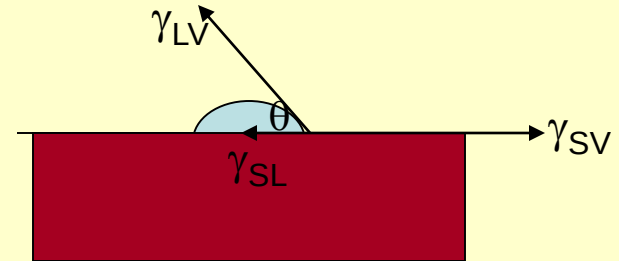
- *Neumann's Equation-of-State* (J. Colloid

Interface Sci. **148** (1992) 190). We may use the

Neumann's empirical equation to determine γ_{SV} .

This equation is valid for Surface Stresses smaller than 72mJ/m² (or 0.072N/m).

$$\cos \theta = 2 \sqrt{\frac{\gamma_{SV}}{\gamma_{LV}}} \exp \left[-\beta (\gamma_{LV} - \gamma_{SV})^2 \right] - 1$$



- *Force Balance Equation* may be used to determine the interfacial stress between the droplet and the surface, γ_{SL} . We can calculate this using the measured surface stress, γ_{SV} , surface tension between the liquid and vapor, γ_{LV} , and the droplet contact angle, θ , parameters in a force balance equation.

$$\gamma_{SL} + \gamma_{LV} \cos(\theta) = \gamma_{SV}$$

Surface Reactions

--- Sublimation

QM is obeyed by sublimed atoms!!!

Sublimation reactions allow atoms to go directly from the solid to gas phase.

The chemical potentials of vapor phase and solid phase must be equal : Condition for sublimation.

$\mu_v = \mu_s$
At low vapor pressure,
where $\mu_v = -kT \ln(kT / p \lambda^3)$
DeBroglie Wavelength $\lambda = h / (2\pi m kT)^{1/2}$
Thus, the equilibrium pressure, p_e is given by

$$p_e = (2\pi m / h^2)^{3/2} (kT)^{5/2} \exp\left(\frac{\mu_s}{kT}\right)$$

Now, we have to select a μ_s . Using a model which assumes harmonic vibrations of frequency, ν , and amplitude equal to the lattice parameter of the solid, we have the free energy of the atom as

$$F / N = \mu_s = U_0 + \left\langle \frac{3}{2} h\nu \right\rangle + 3kT \ln\left(1 - \exp\left(\frac{-h\nu}{kT}\right)\right)$$

We find that at absolute zero, $L_0 = -\left(U_0 + \left\langle \frac{3}{2} h\nu \right\rangle\right)$
 L_0 is the sublimation energy!

At high temperatures, $\ln\left(1 - \exp\left(\frac{-h\nu}{kT}\right)\right) = \ln\left(\frac{h\nu}{kT}\right)$

$$\mu_s = L_0 + 3kT \ln\left(\frac{h\nu}{kT}\right) \quad \text{and} \quad p_e = (2\pi m \nu^2)^{3/2} (kT)^{1/2} \exp\left(\frac{-L_0}{kT}\right)$$

An Arrhenius-type equation

$$p_e T^{-1/2} = \left(\frac{2\pi m \nu^2}{k^3}\right)^{3/2} \exp\left(\frac{-L_0}{kT}\right)$$

Graphical data of the form

$$\text{Log}_{10}(p_e) = A - B/T$$

(since the $T^{-1/2}$ varies relatively slowly, it can be ignored for simplicity)

The constant L_0 is found within B. The constant ν is found within the constant A.

Surface Reactions

--- Crystal Growth from the Vapor Phase

1. Difference between deposition from vapor and sublimation to vapor is in the concept of supersaturation, S . $S=p/p_e$. The change in chemical potential, $\Delta\mu$, is given by: $\Delta\mu=kT \ln S$.
 - **Positive $\Delta\mu$ ($p>p_e$) leads to condensation.**
 - **Negative $\Delta\mu$ ($p<p_e$) leads to sublimation.**
2. Adsorption rate, R_+ , is given by: $R_+=p/(2\pi mkT)^{1/2}$
3. Desorption rate, R_- , is given by: $R_-=v_a \exp(-E_a/kT)$
 v_a is frequency with which atoms leave the surface
4. Diffusion of adatom across surface is given by diffusivity, D :
 $D=(v_d a^2/4 \exp(-E_d/kT))$
5. Lifetime before desorption, τ_a , is given by: $\tau_a=v_a^{-1} \exp(E_a/kT)$
Again, v_a is frequency with which atoms leave the surface
6. Characteristic distance, x , the adatom may diffuse before leaving the surface: $x=(D \tau_a)^{1/2}$

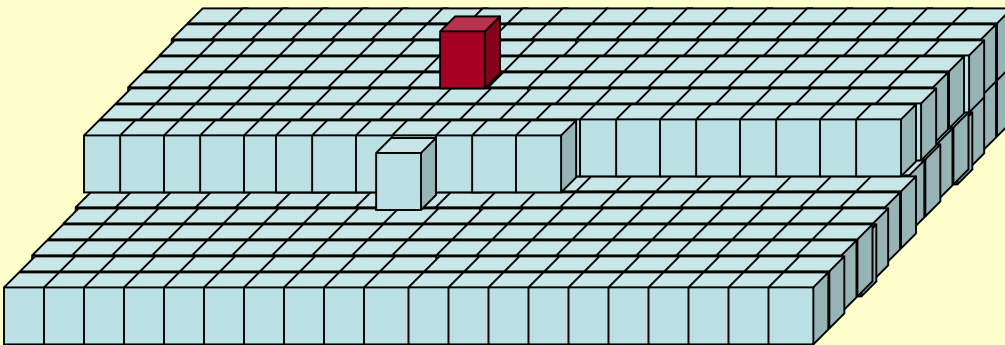
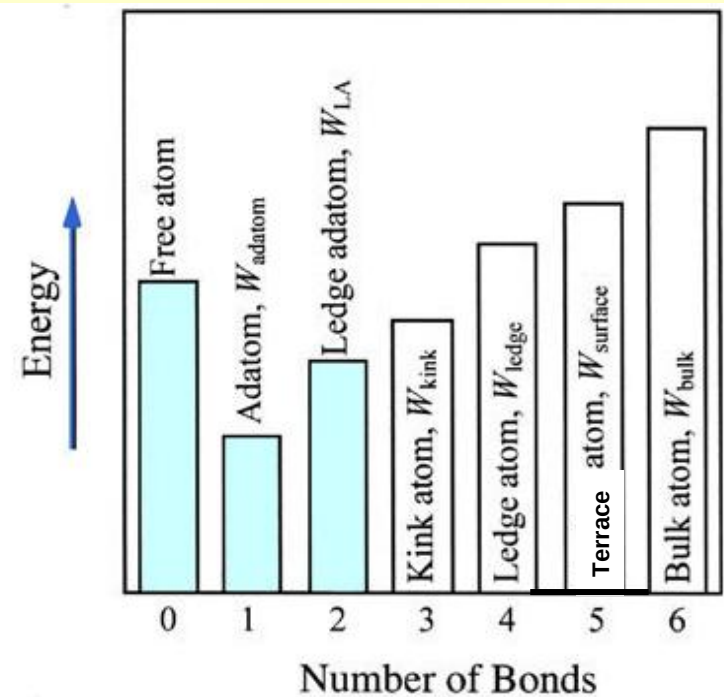
Surface Reactions

--- Crystal Growth from the Vapor Phase

Physical Meaning of x : $x = (D \tau_a)^{1/2}$

Since Site Stability is directly proportionality to binding energy, the adatom has low site stability and will desorb after time τ_a . Given the surface diffusivity, D , the atom must find a more stable site within a distance, x , in order to remain on the surface and lead to growth from the vapor phase.

Binding Energy for Atoms at Various Sites



Summary/Review

What is a Surface? Atom State and Structure

How are Surfaces Classified? Kinks, Terrace, Ledges, Adatoms

Why are surface atoms important? Reactions occur at surface. Additionally, nanostructured material has increased surface to bulk atoms.

What are some of the Properties of surfaces? Thermodynamic Properties (Free Energy, Wulff construction allows surface energy to predict shape of crystal), Density, Surface Tension (or Surface Energy)

What are the broad categories of surface reactions considered? Sublimation and Crystal Growth

Exercise Question 1 (from Venables text)

Consider the (001) face of a fcc crystal. The sublimation energy, L_o , is 3eV and the Einstein frequency factor, ν , is 10THz. Use the appropriate formulations to :

(a) Express the local equilibrium between adatom evaporation, R_- , and the rate of arrival, R_+ , of atoms from the vapor to the surface to find the concentration of adatom monolayer (ML) units.

$$R_+ = R_-$$

$$R_+ = \frac{p}{\sqrt{2\pi mkT}}$$

$$R_- = \nu_a \exp\left(\frac{-E_a}{kT}\right)$$

$$\frac{p}{\sqrt{2\pi mkT}} = \nu_a \exp\left(\frac{-E_a}{kT}\right)$$

$$\frac{p_e}{\sqrt{2\pi mkT}} = \nu_a \exp\left(\frac{-E_a}{kT}\right)$$

$$\frac{p_e T^{-1/2}}{\sqrt{2\pi mk}} = \nu_a \exp\left(\frac{-E_a}{kT}\right)$$

$$p_e T^{-1/2} = \left(\frac{2\pi m \nu^2}{k^3}\right)^{3/2} \exp\left(\frac{-L_o}{kT}\right)$$

$$\frac{\left(\frac{2\pi m \nu^2}{k^3}\right)^{3/2} \exp\left(\frac{-L_o}{kT}\right)}{(2\pi mk)^{1/2}} = \nu_a \exp\left(\frac{-E_a}{kT}\right)$$

$$\frac{(2\pi m \nu)^3}{k} \exp\left(\frac{-L_o}{kT}\right) = \nu_a \exp\left(\frac{-E_a}{kT}\right)$$

$$\frac{(2\pi m \nu)^3}{k} \exp\left(\frac{-L_o}{kT}\right) = \nu_a \exp\left(\frac{-E_a}{kT}\right)$$

Exercise Question 1 (from Venables text)

(b) Find the adatom concentration at 1000K if R = 1ML/sec

$$\frac{(2\pi m \nu)^3}{k} \exp\left(\frac{-L_o}{kT}\right) = \nu_a \exp\left(\frac{-E_a}{kT}\right)$$

L_o is 3eV and the Einstein frequency factor, ν , is 10THz

$$\nu_a \exp\left(\frac{-E_a}{kT}\right) = 1 \text{ ML/sec}$$

To Complete the Problem, Solve for m (mass of adatoms) using the above parameters in appropriate units!!!

Exercise Question 1 (from Venables text)

Consider how might vacancies (e.g. empty lattice sites)- which decrease the Einstein vibrational frequency of neighboring atoms by 80%-- effect adsorbed ML concentration.

By the equation,

$$\frac{(2\pi m\nu)^3}{k} \exp\left(\frac{-L_o}{kT}\right) = \nu_a \exp\left(\frac{-E_a}{kT}\right)$$

Decreasing the vibrational frequency will decrease the sublimation rate (Recall RHS of the equation is R-), Thus decreasing the sublimation rate will increase the adsorption concentration.